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Spatial-Frequency Collaborative Decoupling and Adaptive Gating Network for High-Resolution Remote Sensing Image Semantic Segmentation

Lijian Xie, Bohua Zhao, Aihua Shen, Zhiguo Li, Xin Zhao, Jinsong Deng

***Abstract*—Semantic segmentation of high-resolution remote sensing images faces challenges due to intra-class heterogeneity and inter-class spectral similarity. Traditional single-domain methods struggle to balance computational efficiency, high-frequency details, and a global receptive field. To address this, we propose the Spatial-Frequency Collaborative Decoupling and Adaptive Gating Network (DSFANet). The model employs a parallel CNN-Transformer architecture: in the spatial domain, a lightweight large kernel attention (LightLKA) module captures local high-frequency boundaries; in the frequency domain, a 2D-FFT-based global filter establishes long-range topological dependencies with an $O(Nlog\,N)$ complexity. Additionally, a scene context-aware gating (SCAG) module aggregates global channel-wise statistics to produce two scalar gating weights for fusing the spatial and frequency representations, and a linear-complexity attention fusion block (AFB) bridges the semantic gap between deep and shallow layers. Experiments on four heterogeneous datasets (Vaihingen, Potsdam, LoveDA, and GID) show that DSFANet achieves an optimal balance between accuracy and computational cost. Furthermore, interpretability analysis using Grad-CAM and t-SNE confirms that this dual-domain architecture effectively handles local occlusions, reduces semantic aliasing, and improves inter-class separability. Overall, this work provides an efficient, reliable, and interpretable method for large-scale, fine-grained land use mapping. The code can be obtained here: https://github.com/LijianXie/DSFA-Net.**

***Index Terms*—Semantic segmentation, frequency-domain filtering, adaptive gating mechanism, global context injection, model interpretability.**

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I. INTRODUCTION

ACCURATE Land Use and Land Cover (LULC) monitoring is essential for understanding human-environment interactions, quantifying ecosystem carbon sequestration, and supporting sustainable development [1], [2], [3], [4]. While sub-meter high-resolution earth observation technologies provide a rich data foundation, achieving high-precision, automated semantic segmentation in complex scenes remains a critical challenge [5], [6]. High-resolution imagery is characterized by distinct spatial heterogeneity, containing both highly fragmented fine-grained objects (e.g., isolated rural settlements and narrow road networks) and expansive homogeneous regions (e.g., vast croplands, contiguous forests, and water bodies) [7], [8]. This "intra-class heterogeneity" and "inter-class spectral similarity" expose the representation limitations of existing deep learning models [9].

To tackle these challenges, methods represented by Fully Convolutional Networks (FCN) and Vision Transformers (ViT) have advanced remote sensing interpretation [10], [11], [12]. However, they struggle to resolve the aforementioned dual challenge. Traditional CNN lack global receptive fields, making them highly susceptible to semantic fragmentation and internal voids when dealing with large-scale continuous land features. Conversely, while ViT establish long-range dependencies through self-attention, their quadratic computational complexity, $O(N^2)$, restricts their application on large-sized image tiles [13], [14]. Existing lightweight transformer strategies alleviate computational overhead but inevitably sacrifice crucial high-frequency spatial details, limiting their performance in accurately localizing tiny targets or sharp boundaries [15], [16], [17].

Crucially, the vast majority of current research focuses solely on single "spatial domain" feature mining, ignoring the rich inherent structural information contained in the "frequency domain" of remote sensing images [18], [19], [20]. Geospatial entities often exhibit significant periodic or regular topological distributions. Due to spectral ambiguity or local occlusion, these features are difficult to separate effectively in the spatial domain, yet they show completely different low-frequency or high-frequency response characteristics in the frequency domain [8], [21]. Furthermore, in traditional "encoder-decoder" architectures, dual-branch or multi-scale feature fusion mostly adopts static channel concatenation or element-wise addition [22], [23]. This rigid, non-content-aware fusion method ignores the significant cross-scene

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heterogeneity in high-resolution imagery. Different input patches (e.g., highly fragmented dense urban areas versus homogeneous macroscopic water bodies) possess distinct holistic domain representation preferences. This lack of an adaptive scene-level routing mechanism is a key factor leading to blurred boundaries and semantic aliasing in complex geographic scenes [24], [25], [26].

Based on the above analysis, to overcome the limitations of single-domain feature extraction and static fusion, this study proposes the Dual Spatial-Frequency Attention Network. DSFANet adopts a parallel CNN and Transformer architecture to preserve shallow high-resolution details and model deep semantics through dual-domain collaborative learning. Specifically:

1) By incorporating a learnable 2D-FFT global filter, we construct an $O(NlogN)$ frequency-domain guidance mechanism that efficiently establishes long-range dependencies, effectively resolving large-scale spectral confusion and fine-grained geometric degradation in high-resolution remote sensing imagery.
2) We innovatively proposed the SCAG module and the linear complexity AFB, achieving the adaptive alignment and representation preference calibration of spatial-frequency dual-domain features.
3) Combined with the deep supervision strategy with hierarchically decaying supervisory weights, this mechanism effectively bridges the semantic gap between deep and shallow features, and significantly suppresses background noise and the vanishing gradient problem in deep networks.
4) Experimental validation on four remote sensing datasets (Vaihingen, Potsdam, LoveDA, GID) shows that DSFANet achieves a segmentation accuracy that surpasses existing comparison methods, and reaches the optimal trade-off.

II. Related Work

A. Limitations and Evolution of Spatial Context Modeling

Local feature extraction via CNN drove early progress in high-resolution remote sensing interpretation [27], [28]. To address drastic scale variations and intra-class heterogeneity, researchers advanced local boundary preservation through multi-scale strategies, such as atrous spatial pyramid pooling and multi-resolution parallel branches [29]. However, standard convolution kernels are inherently restricted by their local receptive fields and Euclidean grid constraints [30], [31]. This inductive bias limits the capability to capture macroscopic context across long physical distances, frequently leading to semantic fragmentation and spatial discontinuity in large-scale homogeneous regions, such as continuous forests or extensive water bodies [32], [33]. Consequently, reliance on local neighborhoods falls short of the stringent requirements for global spatial coherence in high-precision LULC mapping [34].

B. Efficiency Bottlenecks of ViT and Large Kernel Convolution Mechanisms

To overcome CNN receptive field constraints, ViT introduced self-attention for global context perception, significantly improving cross-regional semantic consistency [35], [36]. However, the $O(N^2)$ computational complexity of standard self-attention restricts its application on high-resolution imagery [37], [38]. Compromise strategies, such as local window attention or spatial pyramid reduction, alleviate computational overhead but inevitably truncate the global information flow, reintroducing fragmentation in large-scale features [39], [40]. Furthermore, deep Transformers frequently suffer from "feature oversmoothing," where homogenized attention weights obscure small targets and high-frequency boundaries [41]. Alternatively, Large Kernel Attention (LKA) mechanisms within CNN expand receptive fields to simulate global perception [42], [43]. Yet, standard LKA retains redundant dense convolutions with low parameter efficiency. To address this, our work integrates the extremely lightweight LightLKA module, which relies solely on Dilated Depth-wise Convolution to efficiently expand the effective receptive field while preserving high-frequency geometric priors [44].

C. The Absence and Introduction of Frequency-Domain Feature Analysis

The majority of deep learning models for remote sensing rely exclusively on spatial domain feature mining, overlooking the rich structural information inherent in the frequency domain [45], [46]. In complex geographical scenes, spatial domain extraction frequently suffers from spectral ambiguity and local occlusion, especially for regular urban road networks and large homogeneous farmlands [47], [48]. Conversely, frequency domain analysis explicitly separates these topological structures based on low- or high-frequency responses across physical occlusions. Research by Rao et al. confirms that incorporating Fast Fourier Transform (FFT) mechanisms efficiently modulates global features and establishes long-range topological dependencies [49]. Similarly, Liu et al. confirmed that applying FFT significantly improves the extraction of fragmented road networks [50].

In addition, research by Zhang et al. also confirms that introducing a global Fourier module can significantly improve the model's ability to capture macroscopic semantics in heterogeneous remote sensing scenes [51]. Building on these findings, this study incorporates a 2D-FFT global filter with $O(Nlog\,N)$ complexity to explicitly leverage these frequency-domain spatial distributions.

D. Multi Scale Fusion and Adaptive Strategies for Addressing the Semantic Gap

Bridging the "semantic gap" between deep abstract semantics and shallow spatial details remains a critical challenge in decoding high-resolution mapping features [52], [53]. Existing architectures predominantly adopt static channel concatenation or element-wise addition for cross-layer fusion [54]. However, this rigid, non-content-aware approach ignores the specific representation dependencies of different land cover types. For instance, fragmented linear features rely on spatial high-frequency boundaries, whereas expansive water

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bodies depend more on frequency-domain global consistency [24], [25], [26]. Consequently, static fusion fails to dynamically align features and often reintroduces shallow background noise, causing semantic aliasing and boundary artifacts [55], [56]. To solve this, our network deploys a SCAG module for dynamic spatial-frequency routing, coupled with a linear-complexity AFB utilizing deep semantic supervision [57]. This content-aware routing strategy filters redundant noise and ensures coherence during dual-domain feature fusion.

III. Proposed Methods

A. Overview of the Proposed Approach

In the land use/land cover semantic segmentation task of high-resolution remote sensing imagery, models often face two major challenges: first, "intra-class inconsistency," that is, large-area homogeneous ground objects (such as vast farmlands and contiguous water bodies) tend to exhibit texture differences from a local perspective, resulting in internal voids in the prediction results; second, "inter-class semantic confusion" that is, complex artificial ground objects (such as fragmented road networks and dense building clusters) have broken edges and are prone to topological distortion due to interference from shadows or background noise. To address the aforementioned feature representation bottleneck in which "local geometric fineness" and "global semantic consistency" are difficult to balance, this study proposes a deeply supervised dual spatial-frequency attention network. As shown in Fig. 1, DSFANet adopts an asymmetric dual-branch hybrid encoder to extract features. Given an input image $I \in \mathbb{R}^{C\times H\times W}$, the CNN spatial branch is dedicated to capturing high-frequency geometric priors (such as physical boundaries) in shallow layers, while the Global Spectral-Spatial (GSS) branch, through parallel spatial-domain large-kernel convolution and frequency-domain global filtering, achieves joint modeling of large-scale context and periodic structures at an efficient computational cost.

In the decoding stage, to solve the "semantic gap" between deep and shallow layer features, the network innovatively introduces a decoder that includes global context injection and progressive linear attention fusion. Then, through a hybrid loss function and an asymmetric deep supervision strategy, DSFANet achieves fine-grained interpretation of complex heterogeneous landscapes without incurring any additional inference overhead.

B. Dual-Branch Hybrid Encoder

1) Hierarchical CNN spatial branch

The pure Transformer architecture, due to the patch embedding mode adopted in the early stage, would lead to the loss of sub-pixel-level edge features in high-resolution remote sensing images. To address this, this study retains a lightweight hierarchical CNN encoder as the local spatial branch. This branch consists of three consecutive down-sampling stages. Each stage cascades a standard 3×3 convolution, batch normalization (BatchNorm), ReLU activation, and the LightLKA module, and employs 2×2 max-pooling to progressively reduce the spatial dimensions. This branch ultimately outputs multi-scale skip-connection features with channel numbers [64, 128, 256] and resolutions $[\frac{H}{2}\times\frac{W}{2}, \frac{H}{4}\times\frac{W}{4}, \frac{H}{8}\times\frac{W}{8}]$, respectively, providing comprehensive local high-frequency geometric priors for subsequent decoder reconstruction of ground object contours.

2) Dual-Domain Global Spectral-Spatial Branch

The GSS branch first maps the image into a sequence of dimension $C_{trans} = 192$ and inputs it into a dual-domain attention network with a depth of $L = 4$. When processing high-resolution remote sensing features, the standard Self-Attention mechanism often leads to severe memory overflow due to its $O(N^2)$ complexity. To solve this computational bottleneck and achieve better representation capabilities, this study abandons traditional self-attention in each Block and instead designs a parallel dual-stream extraction architecture of the spatial domain and frequency domain.

Specifically, the GSS branch first partitions the input image into non-overlapping patches via a Patch Embedding layer to generate a token sequence of dimension C_{trans}, which is subsequently processed by a stack of L dual-domain blocks. When processing high-resolution remote sensing features, the standard Self-Attention mechanism often leads to severe memory overflow due to its quadratic computational complexity. To resolve this computational bottleneck while retaining robust global representation capabilities, we fundamentally redesign the standard Transformer block. Specifically, the computationally expensive self-attention module is entirely replaced by integrated components. First, in the spatial domain, we introduce the Light Large Kernel Attention (LightLKA, Section III-C) module to efficiently establish mid-to-long-range spatial dependencies with linear complexity, complementing the local high-frequency geometric priors supplied by the CNN branch. In parallel, within the frequency domain, we introduce a 2D Fast Fourier Transform-based Global Filter (Section III-D) to efficiently capture periodic global structures and enhance low-frequency semantic consistency. Finally, to integrate these parallel processing streams, we replace the standard static connections with the Scene-Context-Aware Gating (Section III-E) mechanism, which dynamically aggregates the complementary features from both domains based on the characteristics of each input patch. This modified parallel dual-stream architecture ensures efficient and comprehensive global context modeling while completely circumventing the quadratic memory overhead typical of standard Transformers.

C. Details of Light Large Kernel Convolution

To address the boundary blurring and fine-grained noise problems caused by "intra-class heterogeneity and inter-class similarity" of ground objects in high-resolution remote sensing imagery, this study constructs a LightLKA in the GSS branch. As shown in Fig. 2, LightLKA strategically removes the initial local convolution in the standard LKA, and only utilizes dilated depth-wise convolution to construct long-range dependencies. The LightLKA module aims to establish long-range dependency relationships at an extremely low parameter cost. Specifically, we simplify and decompose the traditional

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large-sized convolutional kernel into two core components: depth-wise dilated convolution (DW-D-Conv) for spatial context aggregation, and point-wise convolution (PW-Conv, i.e., 1×1 convolution) for cross-channel feature interaction. In land use interpretation tasks, DW-D-Conv greatly expands the effective receptive field through the dilation mechanism, enabling efficient capture of spatial long-range dependencies of large-scale continuous ground objects; PW-Conv focuses on linear projection in the channel dimension, deeply mining and fusing high-order spectral attributes of features, thereby enhancing the discrimination ability for complex land classes with subtle spectral differences. The LightLKA module can be written as:

$$F_{Lightlka} = PW - Conv(DW - D - Conv(F)) \otimes F \quad (1)$$

Here, $F \in \mathbb{R}^{C \times H \times W}$ is the input feature, $F_{Lightlka} \in \mathbb{R}^{C \times H \times W}$ denotes the output feature. $\otimes$ means element-wise product.

D. Details of Global Filter

To address the limitation of the quadratic computational complexity of the standard self-attention mechanism, this study introduces a global filtering module based on the 2D Fast Fourier Transform into the GSS branch. As shown in Fig. 2, this module orthogonally transforms spatial features into the complex frequency spectrum domain, uses a learnable complex weight matrix for global dynamic modulation, and then reconstructs them back to the spatial domain through inverse transformation. This frequency-domain mechanism establishes full-image context dependencies with an efficient complexity of $O(NlogN)$, which can accurately capture the periodic structural features of geographic entities such as complex land features (e.g., farmlands and building clusters). Meanwhile, by adaptively calibrating frequency components, the module dynamically reinforces low-frequency global consistency. This complements the high-frequency boundary extraction in the spatial domain, significantly improving intra-class uniformity for large homogeneous regions (e.g., open water bodies and forested areas).

Given the input feature $\boldsymbol{X} \in \mathbb{R}^{C \times H \times W}$, 2D rFFT is first performed along the spatial dimensions to transform it from the spatial domain to the frequency domain. The 2D discrete fourier transform (DFT) is defined as follows:

$$\mathcal{X}[u,v] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} \boldsymbol{X}[m,n]\, e^{-j2\pi(\frac{um}{H}+\frac{vn}{W})} \quad (2)$$

$\mathcal{X}[u,v]$ denotes the complex spectrum, where u, v are frequency domain coordinates, m, n are spatial coordinates. The input feature $\boldsymbol{X}$ is a real tensor whose frequency domain response satisfies conjugate symmetry, i.e. $\mathcal{X}[H-u, W-v] = \mathcal{X}^*[u,v]$,($*$ denotes complex conjugation). To eliminate redundant computations and significantly reduce the number of parameters, we adopt the real-valued Fast Fourier Transform (rFFT) in our implementation, retaining only half of the spectral features to obtain $\mathcal{X}_r \in \mathbb{C}^{C \times H \times W'}$,where $W' = \left\lfloor \frac{W}{2} \right\rfloor + 1$. Subsequently, we introduce a learnable global complex weight matrix $\boldsymbol{K} \in \mathbb{C}^{C \times H \times W'}$ to adaptively modulate the spectrum within the frequency domain. This learnable complex weight matrix $\boldsymbol{K}$ can be interpreted as a data-driven spectral gating mechanism. Unlike fixed hand-crafted filters, $\boldsymbol{K}$ is optimized end-to-end to automatically reweight different frequency components according to their relevance to the semantic segmentation task. Concretely, $\boldsymbol{K}$ is expected to learn to enhance low-frequency components that capture global shape and category-level appearance consistency, which is crucial for identifying large homogeneous regions (e.g., water bodies, farmland) in remote sensing imagery; Simultaneously, it selectively preserves high-frequency components corresponding to sharp boundaries and fine textures (e.g., building edges) to prevent over-smoothing. Furthermore, it acts to suppress mid-to-high frequency components that primarily stem from sensor noise, atmospheric scattering, or irrelevant textural variations, thereby improving the model's robustness across diverse remote sensing scenes. This process is realized through the Hadamard product:

$$\boldsymbol{y_r} = \boldsymbol{K} \odot \mathcal{X}_r \quad (3)$$

Finally, we reconstruct the modulated complex spectrum $\boldsymbol{y_r}$ back to the spatial domain through the 2D inverse real Fast Fourier Transform (2D irFFT) to obtain the updated feature $\boldsymbol{Y} \in \mathbb{R}^{C \times H \times W}$. The 2D inverse transform formula is as follows:

$$\boldsymbol{Y}[m,n] = \frac{1}{HW} \sum_{u=0}^{H-1} \sum_{v=0}^{W-1} \boldsymbol{y}[u,v]\, e^{+j2\pi(\frac{um}{H}+\frac{vn}{W})} \quad (4)$$

E. Scene-Context-Aware Gating for Dual-Domain Fusion

In the parallel dual-domain feature extraction process, the spatial domain tends to capture local geometric boundaries and high-frequency textures, while the frequency domain is adept at establishing the low-frequency consistency and periodic global structures of large-scale homogeneous regions. However, high-resolution remote sensing images exhibit significant cross-scene heterogeneity, meaning different input patches possess distinct domain representation preferences. As shown in Fig. 2, to avoid the suboptimal representations caused by static feature concatenation, this study designs a SCAG mechanism. Instead of performing pixel-level independent routing, SCAG aggregates global channel-wise statistics to produce two scalar gating weights for fusing the spatial and frequency representations. Given the normalized input feature $\boldsymbol{Y} \in \mathbb{R}^{C \times H \times W}$ of the dual-domain module, it is first compressed into a channel-level context descriptor $z \in \mathbb{R}^{C \times 1 \times 1}$ through spatial Global Average Pooling:

$$z = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{Y}_{i,j} \quad (5)$$

Subsequently, to capture the cross-dimensional nonlinear dependencies between channels, z is input into a bottleneck network containing two layers of 1×1 convolutions. The network first reduces the dimension to C/r (with a reduction ratio $r = 16$ to balance the computational overhead and the

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channel representational capacity) to filter out redundancy, and then projects it into two one-dimensional feature components corresponding to the dual branches. Finally, through Sigmoid function activation, two independent scene-level gating scalars α and β with a value range between (0, 1) are solved:

$$[\alpha, \beta]^T = SCAG(Y) = \sigma(W_2\delta(W_1 z)) \quad (6)$$

where $W_1 \in \mathbb{R}^{\frac{c}{r}\times C}$ and $W_2 \in \mathbb{R}^{2\times\frac{c}{r}}$ represent the convolutional weight matrices for dimensionality reduction and expansion, respectively, δ denotes the ReLU activation function, and σ denotes the Sigmoid function. The calculated gating scalars subsequently perform feature weighting on the spatial domain output F_{spa} and the frequency domain output F_{freq} through a broadcasting mechanism, respectively, and establish a residual connection with the original input $\boldsymbol{X}$:

$$F_{out} = X + DropPath(\alpha \cdot F_{spa} + \beta \cdot F_{freq}) \quad (7)$$

$$F_{spa} = \mathcal{F}_{Lightlka}(Y), F_{freq} = \mathcal{F}_{GF}(Y) \quad (8)$$

It is worth emphasizing that this module replaces the conventional softmax normalization with independent Sigmoid activations. The design motivation is to decouple the scaling of the dual-branch features, thereby providing the network with the architectural flexibility to independently enhance both components. By adopting independent Sigmoid gates, the scalars α and β function as independent feature enhancement weights directly optimized by the loss function. Consequently, the network can adaptively assign higher gating weights to the branch that exhibits stronger representational capacity and extracts more accurate features for the current input. This gating strategy fosters a synergistic fusion, ensuring the lossless coexistence of local geometric fineness and global context.

F. Progressive Attention-Guided Decoder

In high-resolution remote sensing land use mapping, the core task of the decoder is to gradually restore the deep, highly abstract semantic features to the original spatial resolution. However, although shallow features are rich in fine spatial geometric boundaries, they inevitably contain a large amount of high-frequency background noise; while deep features have strong semantic representation but blurred spatial localization. Traditional decoders usually adopt channel concatenation. This fusion method is difficult to solve the "semantic gap" between the two, and easily leads to "jagged" edges of complex land features and inter-class feature overlap phenomena. As shown in Fig. 2, to solve this problem, this study designs a progressive attention-guided decoder. This decoder is composed of sub-pixel upsampling and an AFB cascaded alternately.

First, to avoid the checkerboard artifacts that may be caused by conventional transposed convolutions, the network adopts the pixelshuffle operator for upsampling. Given the deep feature F_{dec}^{i-1} of the $i-1$ stage, by reorganizing the mapping relationship between the channel and spatial dimensions, it is losslessly upsampled to the same spatial resolution as the skip-connection feature F_{cnn}^{i} of the i layer CNN encoder to obtain $G^i \in \mathbb{R}^{C\times H\times W}$.

Subsequently, to overcome the semantic ambiguity brought by direct concatenation, the AFB module is introduced. Specifically, G^i and F_{cnn}^{i} first undergo channel projection and alignment through 1×1 convolutions, respectively, and are then concatenated in the channel dimension and fused for local and cross-scale interactions via a 3×3 convolutional layer to obtain the initial fused feature F_{fuse}^{i}. On this basis, since the standard self-attention mechanism on high-resolution feature maps faces the memory overflow problem of $O(N^2)$, this study innovatively introduces an efficient self-attention mechanism with a linear complexity of $O(N)$ at the end of the fusion to enhance feature representation.

During the specific calculation process, the feature F_{fuse}^{i} is linearly projected into Q, K, and V matrices. Distinct from computing a massive spatial similarity matrix, this mechanism first performs L2 normalization on Q and K, then preferentially calculates the matrix product of K^T and V to aggregate and generate a Global Context Descriptor, and finally multiplies it with Q and applies a learnable scaling parameter γ to obtain the attention residual output. $\phi(\cdot)$ denotes the L2 normalization operation along the channel dimension. The entire decoding fusion process can be expressed as:

$$F_{align}^{i} = Concat(Proj(G^i), Proj(F_{cnn}^{i})) \quad (9)$$

$$F_{fuse}^{i} = \delta(BN(Conv_{3\times3}(F_{align}^{i}))) \quad (10)$$

$$F_{dec}^{i} = F_{fuse}^{i} + \gamma(\phi(Q)(\phi(K)^T V)) \quad (11)$$

Through the aforementioned linear attention guidance, the AFB module can fully utilize the global semantics contained in the deep features, dynamically filter the invalid background noise in the shallow CNN features, and then selectively enhance and sharpen the edge features.

G. Hybrid Objective Function and Deep Supervision Strategy

In high-resolution remote sensing land use semantic segmentation tasks, complex and variable scenes are usually accompanied by severe class distribution imbalance problems. To ensure that the model can achieve high-fidelity predictions on both dominant and rare categories, the main supervision branch of this study adopts a hybrid objective function, namely the combined form of Cross-Entropy Loss ($\mathcal{L}_{ce}$) and Dice Loss ($\mathcal{L}_{dice}$):

$$\mathcal{L}_{main} = \mathcal{L}_{ce}(P_{main}, y) + \mathcal{L}_{dice}(P_{main}, y) \quad (12)$$

where P_{main} denote the predictions from the main branch, and y represents the ground truth.

Furthermore, given that deep networks are prone to facing the dilemma of gradient vanishing during the backpropagation process, this study introduces a Deep Supervision ($\boldsymbol{\mathcal{L}_{ds}}$) strategy. Specifically, we derive two auxiliary prediction heads at the deep and intermediate scale stages of the decoder, respectively. During the forward propagation of training, these two auxiliary prediction features are precisely upsampled to the full spatial resolution of the original image through

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Bilinear Interpolation, and then the auxiliary classification loss is calculated.

Notably, the auxiliary supervision signals are only applied to the deep and intermediate decoding stages containing highly abstract semantics, and deliberately avoid the shallowest decoding stage (i.e., AFB3). This asymmetric design follows the objective law of hierarchical feature learning of the network: the main task of the deep network is to achieve efficient determination of semantic categories, while the shallow network needs to focus on refinement of spatial geometric boundaries. If an additional classification loss is forcibly applied to the shallowest layer, it is very easy to force the shallow features to degrade towards overly smooth semantic clustering, thereby damaging the core ability of the model to reconstruct high-frequency edges.

Algorithm 1 Forward Pass and Deep Supervision Training of DSFANet

Input: Image $I \in \mathbb{R}^{C\times H\times W}$,Label $\boldsymbol{Y}$ (training only), Depth $\boldsymbol{L}$
Output: Predicted map P_{main} (Inference) or Total loss $\mathcal{L}_{total}$
/ Phase 1: Dual-Branch Hybrid Encoding /
$F_{cnn}^{0} \leftarrow \boldsymbol{I}$
for i =1,2,3 **do**
  $F_{cnn}^{i} \leftarrow MaxPool(LightLKA(ConvBlock(F_{cnn}^{i-1})))$ // Local spatial features
end for
$\boldsymbol{X} \leftarrow PatchEmbedding(\boldsymbol{I})$ // Sequence mapping
for l = 1 **to** $\boldsymbol{L}$ **do**
  $X_{norm} \leftarrow BatchNorm(\boldsymbol{X})$
  $F_{spa}, F_{freq} \leftarrow LightLKA(X_{norm}), irFFT2(\boldsymbol{K} \odot rFFT2(X_{norm}))$
  $\alpha, \beta \leftarrow SCAG(X_{norm})$ // Scene-Context-Aware Gating
  $\boldsymbol{X} \leftarrow \boldsymbol{X} + DropPath(\alpha \cdot F_{spa} + \beta \cdot F_{freq})$ // Adaptive dual-domain fusion
  $\boldsymbol{X} \leftarrow \boldsymbol{X} + MLP(BatchNorm(\boldsymbol{X}))$
end for
$\boldsymbol{X} \leftarrow \boldsymbol{X} + GlobalContextPool(\boldsymbol{X})$ // Residual macro-context injection
/ Phase 2: Progressive Attention-Guided Decoding /
G ← $\boldsymbol{X}$
for j = 1, 2, 3 **do**
  $G \leftarrow AFB(PixelShuffle(G), F_{cnn}^{4-j})$ // O(N) linear attention integration
  if Training **and** $j \leq 2$ **then** $P_{aux}^{j} \leftarrow \text{AuxHead}_j(G)$ // Deep supervision
end for
$P_{main} \leftarrow \text{MainHead}(\text{PixelShuffle}(G))$
/ Phase 3: Hybrid Objective Function /
if not Training **then return** P_{main} // Zero inference overhead
$P_{main}, P_{aux}^{1}, P_{aux}^{2} \leftarrow \text{Bilinear Interpolate}(\ldots, (\text{H}, \text{W}))$
$\mathcal{L}_{total} \leftarrow \mathcal{L}_{ce_and_dice}(P_{main}, y) + \sum_{j=1}^{2} \lambda_j \mathcal{L}_{ce}(P_{aux}^{j}, y)$
return $\mathcal{L}_{total}$

Therefore, during the training phase, the total loss function $\mathcal{L}_{total}$ is defined as the weighted sum of the main branch loss and the auxiliary branch losses:

$$\mathcal{L}_{total} = \mathcal{L}_{main} + \lambda_1 \mathcal{L}_{ce}(P_{aux1}, y) + \lambda_2 \mathcal{L}_{ce}(P_{aux2}, y) \quad (13)$$

$$\mathcal{L}_{ce} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \quad (14)$$

$$\mathcal{L}_{dice} = 1 - \frac{2}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} \frac{\hat{y}_k^{(n)} y_k^{(n)}}{\hat{y}_k^{(n)} + y_k^{(n)}} \quad (15)$$

where P_{aux1} and P_{aux2} denote the predictions from the two auxiliary branches, respectively. The balancing weights are empirically set to $\lambda_1 = 0.4$ and $\lambda_2 = 0.2$ [58], [59]. where N and K represent the number of samples and the number of classes, respectively. $y^{(n)}$ and $\hat{y}^{(n)}$ represent the one-hot encoding of the true semantic labels and the corresponding softmax output of the network, $n \in [1, \ldots, N]$. $\hat{y}_k^{(n)}$ is the confidence of sample n belonging to the category k. During the inference phase, all auxiliary prediction branches will be completely discarded, and the network only retains the main branch for fast forward propagation. The learning process is detailed in Algorithm 1.

IV. Experiments and Results

A. Datasets

To verify the analytical robustness of the proposed architecture under different spatial resolutions and heterogeneous geographical morphologies, this study conducted evaluations on four representative large-scale public remote sensing benchmark datasets: ISPRS Potsdam, ISPRS Vaihingen, LoveDA, and GID.

1) The ISPRS Potsdam and Vaihingen datasets are standard semantic annotation benchmarks provided by the International Society for Photogrammetry and Remote Sensing (ISPRS). These two datasets focus on the parsing of complex urban morphologies at extremely high resolutions [60]. Among them, the Potsdam dataset contains 38 true orthophoto images with a size of 6000 × 6000 pixels and a spatial resolution of 5 cm; while the Vaihingen dataset contains 33 images with a resolution of 9 cm. Both datasets finely divide the land surface into six semantic categories: impervious surfaces, building, low vegetation, tree, car, and clutter/background. These two datasets exhibit extremely high intra-class variance and severe shadow occlusion, and are therefore used to verify the representation capabilities of the feature encoder and progressive attention decoder in this study for extracting fine-grained geometric edges (such as vehicle contours and building boundaries). In quantitative evaluation, we excluded the "clutter/background" category, but retained it for qualitative visualization mapping [61], [62].

2) The LoveDA dataset contains 5987 high spatial resolution (0.3 m) remote sensing images of 1024 × 1024 pixels and RGB spectral bands. It is separated into two parts: urban and rural, and each part consists of the training set, validation set, and test set. The LoveDA dataset contains seven categories: background, building, road, water, barren, forest, and agriculture [63]. Since the test set's data from the LoveDA dataset is not annotated, the prediction results need to be uploaded to the specified website to obtain the results, but

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the website only gives the IoU and mIoU metrics. This is not sufficient to evaluate the model. Specifically, utilizing the publicly available labeled images, we employed an independent random sampling strategy to split the official LoveDA validation set into validation set and test set [64].

3) The Gaofen Image Dataset (GID) is introduced to test the model's LULC mapping capability under large-scale satellite observations [65]. This dataset contains 150 multispectral images with broad spatial coverage (single image dimensions up to 6800 × 7200) and a spatial resolution of 1 m. The experiment adopted its standard 5-class classification system: built-up, farmland, forest, meadow, and water. Different from the fragmentation of urban datasets, the prominent feature of GID is the continuous distribution of large-area homogeneous regions and blurred natural transition boundaries. This characteristic can directly and objectively verify the effectiveness of the 2D real Fast Fourier Transform global filtering module introduced in the GSS branch in this study in capturing long-range periodic structures and enhancing low-frequency semantic consistency.

B. Experimental Implementation Details

To ensure the fairness of the evaluation and the reproducibility of the results, the training and testing of all network models were conducted under a unified hardware platform and software framework. The specific hardware configuration is a high-performance graphics workstation equipped with an Intel Core Ultra 9 285K CPU, 96 GB RAM, and an ASUS GeForce RTX 5080 TUF OC 16GB GPU. In terms of network optimization, all experiments adopted the Adam optimizer to perform gradient descent and weight updates, and the initial learning rate was uniformly set to 0.001. Limited by the huge computational overhead brought by high-resolution remote sensing images and the objective physical constraints of GPU memory, the batch size for model training was set to 4. The number of epochs for the entire training process was set to 100 to ensure that the network converged to a stable state. To objectively evaluate the generalization ability of the model, this study strictly divided all datasets into non-overlapping training, validation, and test sets according to a ratio of approximately 7:2:1. In this process, a Random Shuffling strategy was adopted for independent sampling [66].

To comprehensively and objectively verify the advancement of the proposed DSFANet in high-resolution remote sensing land cover parsing tasks, this study selected nine highly representative state-of-the-art semantic segmentation networks as comparison baselines, specifically including: ABCNet [57], BANet [67], CMLFormer [68], CMTFNet [69], MANet [70], SFFNet [62], AFENet [71], MIFNet [72], SFCVPNet [73].

These baseline models cover diverse deep learning paradigms (i.e., pure CNN architectures, pure Transformer architectures, and CNN-Transformer heterogeneous hybrid architectures). By comparing with these advanced methods under the same experimental conditions, it aims to demonstrate from multiple dimensions such as parameter efficiency, local boundary fidelity, and global semantic consistency, the effectiveness and superiority of the dual-domain fusion and progressive attention decoding strategies designed in this study.

C. Evaluation Metrics

To evaluate the performance of the models, the Overall Accuracy (OA), mean F1 score (mF1), the mean Intersection over Union (mIoU), and Kappa coefficient are used as evaluation metrics.

$$OA = \frac{\sum_{k=1}^{K} TP_k}{N} \tag{16}$$

$$Kappa = \frac{OA - P_e}{1 - P_e} \tag{17}$$

$$P_e = \frac{1}{N^2}\sum_{k=1}^{K}((TP_k + FP_k) \times (TP_k + FN_k)) \tag{18}$$

$$Precision_k = \frac{TP_k}{TP_k + FP_k}, Recall_k = \frac{TP_k}{TP_k + FN_k} \tag{19}$$

$$F1_k = \frac{2 \times Precision_k \times Recall_k}{Precision_k + Recall_k}, mF1 = \frac{1}{K}\sum_{k=1}^{K} F1_k \tag{20}$$

$$IoU_k = \frac{TP_k}{TP_k + FP_k + FN_k}, \text{mIoU} = \frac{1}{K}\sum_{k=1}^{K} IoU_k \tag{21}$$

Where, K denotes the number of classes, N denotes the total number of pixels in the image. For any class $k \in \{1,2,\dots,K\}$, we define the pixel counts for its corresponding True Positives (TP_k), False Positives (FP_k), and False Negatives (FN_k).

D. Results on the ISPRS Vaihingen Dataset

The quantitative comparison experimental results in Table I show that DSFANet achieved the optimal performance on all comprehensive metrics, with an OA of 91.28%, an mF1 of 89.83%, an mIoU of 81.68%, and a Kappa of 0.8837. Compared to the comparison methods, our method shows a clear advantage over the suboptimal method CMTFNet, exceeding it by 0.84% and 1.10% in OA and mIoU, respectively, which indicates that the proposed dual-scale fusion strategy effectively enhances the feature expression ability in high-resolution remote sensing scenes. In terms of per-class performance, DSFANet effectively extracts artificial objects and natural vegetation. Specifically, it achieves the highest IoU and F1-scores in the “Building,” “Low.veg,” “Imp. surf.,” and “Tree” categories. Notably, the “Low.veg” category is evident: DSFANet yields an IoU of 76.91%, outperforming the second-best method (ABCNet) by 2.21%. Although CMTFNet performs slightly better in the "Car" category (IoU of 77.09% compared to 74.95% for DSFANet), DSFANet maintains balanced and stable results across most categories without significant fluctuations. This confirms the reliability and consistency of the proposed method in complex scenarios.

The qualitative results in Fig. 3 show that the segmentation maps generated by DSFANet are qualitatively closest to the ground truth in various complex urban scenes, showing the

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best visual effects. Specifically, it effectively preserves the sharp boundaries and coherent internal structures of artificial objects, especially for the "buildings" category, significantly reducing fragmentation phenomena. In addition, in regions with high spectral similarity, such as intricately mixed areas of "Low.veg" and "Tree", DSFANet shows stronger discrimination ability, greatly reducing the misclassifications in CMLFormer and ABCNet, which echoes the high quantitative metrics reported in Table I.

E. Results on the ISPRS Potsdam Dataset

Table II shows the quantitative comparison results, in which the proposed DSFANet generally outperforms the other nine advanced methods. In terms of overall metrics, DSFANet achieved the highest scores on all data, recording an OA of 89.31%, an mF1 of 88.11%, an mIoU of 78.90%, and a Kappa of 0.8603. Among them, our method is 1.70% higher in mIoU and 0.86% higher in OA than the strong competitor CMTFNet, further verifying the robustness of the dual-scale fusion architecture in different urban landscapes. In terms of per-class performance, DSFANet effectively distinguishes categories with complex spectral features and small objects. Specifically, it achieves the highest IoU and F1-scores in the "Imp. surf.", "Low.veg", "Tree", and "Car" categories. The small-object "Car" category is particularly notable: DSFANet outperforms the second-best method, AFENet (70.65%), by 1.78% in IoU, and leads ABCNet (67.30%) by a larger margin. Although methods like MANet score slightly higher in the "Building" category (with an IoU of 85.98% compared to 84.35% for DSFANet), DSFANet maintains a more balanced and stable performance distribution across all categories.

Fig. 4 shows the qualitative segmentation results of DSFANet. As shown in the Fig. 4, in various complex urban scenes, the segmentation maps generated by DSFANet are visually closest to the ground truth labels. Methods like BANet and CMLFormer usually show obvious noise and misclassifications in complex areas. In comparison, our model shows excellent ability in maintaining the geometric integrity of artificial features; specifically, it retains sharp and coherent boundaries for "Building" and "Imp. surf.", effectively reducing the "salt and pepper" noise common in other methods. In addition, in areas with high spectral similarity, especially the complex interaction areas between "Low.veg" and "Tree", DSFANet shows stronger discrimination ability, significantly reducing the confusion between vegetation types. It is worth noting that the model also shows robust performance in identifying small targets such as "Car", reducing missed detections compared with SFFNet and ABCNet, further verifying the effectiveness of the model.

It is worth noting that the proposed model exhibits obvious class-level performance differences on the two datasets: in the Vaihingen dataset, the model performs best on the building and low vegetation categories; while in the Potsdam dataset, the model performs excellently on the impervious surface, tree, and car categories. This phenomenon can be attributed to the correlation between the specific physical characteristics of the datasets and the dual-domain extraction mechanism of the network. The Vaihingen dataset is characterized by a spatial resolution of 9 cm, detached houses, high roof contrast, and continuous lawns. For this suburban layout, the frequency-domain global filter can effectively model the low-frequency uniformity of large lawns, while the spatial LightLKA branch can capture the clear geometric boundaries of independent buildings. The SCAG module adaptively balances these features, thereby reducing the inter-class confusion of these specific categories. On the contrary, the 5 cm ultra-high spatial resolution and dense urban morphology of the Potsdam dataset provide richer high-frequency textures for small objects. The depthwise convolution in LightLKA provides a larger receptive field, which can effectively encode the fine local details of cars and trees.

In addition, although Potsdam's concrete flat roofs and impervious surfaces (such as roads) have high spectral similarity, the global filter alleviates local ambiguity by modeling long-range topological dependencies in the frequency domain. Supported by the multi-scale context aggregation of the AFB, the network relies on macroscopic structural priors rather than ambiguous local spectral features, thereby being able to robustly segment impervious surfaces and trees in complex urban landscapes.

F. Results on the LoveDA Dataset

Table III shows the quantitative comparison results on the LoveDA dataset, which is characterized by complex urban-rural scenes. Although this dataset is highly challenging, the proposed DSFANet still achieved the best overall performance among all comparison methods, recording an OA of 59.68%, an mF1 of 63.49%, an mIoU of 49.61%, and a Kappa of 0.5161. Although the lead in mIoU over the suboptimal model ABCNet is relatively weak, DSFANet shows excellent capabilities in specific semantic categories that require unique geometric and context modeling. Specifically, it achieves the highest IoU and F1-scores in the "Background," "Building," and "Road" categories. The improvement in the challenging "Road" category is distinct: DSFANet outperforms the second-best method, ABCNet (48.30%), by 4.11% in IoU, and BANet (42.99%) by 9.42%. This suggests that the proposed method effectively preserves the connectivity of linear features in complex scenes. Although other models perform better in certain natural feature categories (for instance, ABCNet achieves the highest IoU of 78.59% in "Water," MANet leads in "Barren," and SFFNet scores slightly higher in "Forest"), DSFANet maintains a more balanced performance distribution between artificial and natural objects. To further test the reliability of this algorithm on the LoveDA dataset, we conducted experiments using multiple random seeds and report the mean and standard deviation. The reliability of this improvement is demonstrated by providing a statistical significance analysis, which is described in supplementary material Appendix A.

Fig. 5 shows the qualitative segmentation results. This dataset is characterized by complex urban-rural transition scenes. It can be clearly seen from the visualization results that the segmentation maps generated by DSFANet are better in quality. Unlike competing methods such as BANet and CMLFormer, which often struggle to maintain the structural integrity of linear features, DSFANet shows excellent ability in maintaining the connectivity of "Road"; Specifically, it

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effectively mitigates the fragmentation and broken road problems common in other baseline models, ensuring a smoother and more continuous road network. In addition, in dense urban areas, our model performs excellently in maintaining the geometric regularity of "Building", minimizing the adhesion of adjacent structures and reducing boundary noise. Even in scenes with complex land cover mixing, such as areas where "Agriculture" and "Water" are intertwined, DSFANet also shows strong discrimination ability, significantly reducing misclassifications and providing clearer semantic boundaries, which is consistent with the high quantitative performance reported in table III.

G. Results on the GID Dataset

Table IV shows the quantitative comparison results, in which the performance of the proposed DSFANet outperforms the other nine advanced methods. In terms of overall evaluation, DSFANet achieved the highest scores on all comprehensive metrics, recording an OA of 87.09%, an mF1 of 76.07%, an mIoU of 65.66%, and a Kappa coefficient of 0.7971. The results show that our method is 1.55% higher in mIoU and 0.81% higher in OA than the strong competitor CMTFNet, verifying the effectiveness of the proposed architecture in processing large-scale remote sensing images. In terms of per-class performance, DSFANet demonstrates robustness in distinguishing natural features with similar spectral characteristics. Specifically, it achieves the highest IoU and F1-scores in the "Farmland" and the challenging "Meadow" categories. The "Meadow" class is distinct: DSFANet outperforms the second-best method, MANet (20.40%), by 2.78% in IoU and substantially exceeds CMLFormer (4.11%). Although certain models score slightly higher in specific categories (e.g., MANet in "Built-up", MIFNet in "Forest", and AFENet in "Water"), DSFANet maintains a more balanced performance distribution across all classes.

Fig. 6 shows the qualitative segmentation results, where the segmentation maps generated by DSFANet are visually closest to the ground truth labels, showing excellent continuity and less noise across different land cover types. Compared to competing methods such as CMLFormer and BANet, which often suffer from severe "salt and pepper" noise and fragmentation in large homogeneous areas, our model shows better performance in maintaining semantic consistency; specifically, it effectively preserves the continuous geometric shapes of "built-up" areas and the homogeneity of "cropland", minimizing internal holes. In addition, in scenes with severe spectral confusion, especially in the subtle distinction between "meadow" and other vegetation types such as "forests", DSFANet shows stronger discrimination ability. Although other baseline models frequently misclassify "meadow" due to spectral similarity, our method identifies these areas with higher fidelity, thereby confirming the significant quantitative lead of our method in the "Meadow" category in Table IV.

V. DISCUSSION

A. Ablation Experiments

Table V and Fig. 7 detail the progressive ablation study on the Vaihingen dataset. The baseline (Ablation 1), a standard U-shaped CNN model employing PixelShuffle, yields a sub-optimal mIoU of 56.37% and struggles with small-scale targets (e.g., "Car" IoU: 26.81%). Independently integrating the spatial LightLKA (Ablation 2) or the frequency-domain Global Filter (Ablation 3) improves the mIoU to 69.25% and 71.83%, respectively, proving their specific utilities in extracting local fine-grained details and capturing global topological structures. To verify the necessity of the proposed adaptive gating mechanism, we introduce a static fusion control experiment (Ablation 4) using parameter-free element-wise addition to combine the dual-branch features, which achieves 86.42% OA, 84.15% mF1, 73.13% mIoU, and 0.8123 Kappa. Replacing this static fusion with the SCAG adaptive gating mechanism (Ablation 5), while keeping all other settings unchanged, further improves the OA to 87.69%, mF1 to 85.54%, mIoU to 75.07%, and Kappa to 0.8354, and increases the "Car" IoU to 68.16%. The consistent gains over static fusion demonstrate that the benefit of SCAG is not merely due to the availability of dual-domain features, but arises from its adaptive gating for synergistic fusion. Subsequently, introducing the Attention Fusion Block (AFB) into the decoder (Ablation 6) effectively improves the precision of target contours in complex scenes, further raising the mIoU to 79.55%. Finally, incorporating deep supervision (Ablation 7) achieves the optimal 81.68% mIoU for the proposed DSFANet. This optimization ensures smoother gradient propagation and strict semantic consistency, significantly enhancing challenging categories such as "Low.veg" (76.91%) and "Car" (74.95%).

B. Parameter Sensitivity Analysis

To evaluate the robustness of the proposed model architecture, this paper conducts a sensitivity analysis on two core hyperparameters of DSFANet: The GSS branch depth (L) and the feature embedding dimension (C). As shown in Fig. 8, regarding the analysis of the GSS branch depth L, we observed that the model maintained high segmentation accuracy when L=2 and L=4, but when the depth further increased ($L \geq 6$), the performance showed a significant decline. This phenomenon can be attributed to the "over-smoothing" problem common in deep vision Transformers; that is, excessive frequency-domain filtering and attention interaction will smooth out the high-frequency geometric boundary information that is crucial in high-resolution remote sensing images. Although L=2 shows a slight numerical advantage on this metric, a network that is too shallow may lack the hierarchical abstraction ability needed to process extremely complex urban scenes. Setting L=4 sacrifices a negligible amount of accuracy on simpler validation sets but provides significantly stronger hierarchical abstraction and robustness across all four diverse datasets. Therefore, considering the maturity of dual-domain feature interaction and the number of parameters comprehensively, we choose L=4 as the best trade-off point between representation ability

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and avoiding the risk of over-smoothing. In addition, Fig. 8 shows the impact of the feature embedding dimension C on the model capacity. With the increase of dimension C, the mIoU of the model reaches a saturation peak at C=192. Further expanding the dimension to C=256 not only failed to bring performance gains, but instead introduced potential overfitting risks due to the surge in parameters, resulting in a slight decrease in accuracy. Based on the above comprehensive sensitivity evaluation and trade-off analysis, we determine the parameter configuration of DSFANet as L=4 and C=192.

C. Model Robustness Verification Based on Grad-CAM

To deeply analyze the feature representation mechanism of the proposed architecture in complex remote sensing scenes, this paper uses Grad-CAM technology to visually analyze the class activation maps of DSFANet and mainstream comparison models [56], [74]. As shown in Fig. 9, it can be observed that when conventional comparison models process large-scale continuous ground objects such as impervious surfaces (Imp. surf.) and low vegetation (Low. veg.), due to the lack of effective modeling of macro context, their activation regions often show significant fragmentation and response faults, and are highly susceptible to physical interference from tree canopy occlusion or high-rise building shadows. In contrast, the activation response of DSFANet on such homogeneous regions shows extremely high regional coherence and structural integrity. This visual evidence further confirms the original design intention of the frequency-domain global filter: by establishing long-range topological dependencies in the frequency-domain space, the network can effectively span local spatial occlusion and pixel-level noise, and remarkably mitigates the “hole” phenomenon in large-scale ground object prediction in complex urban scenes.

D. t-SNE Visualization Comparison of Deep Feature Spaces Across Different Models

This paper adopts the t-SNE algorithm to conduct two-dimensional dimensionality reduction visualization on the deep semantic features output at the end of the network decoder. This method can intuitively reflect the intra-class compactness and inter-class separability of features in the manifold space, thereby verifying the rationality of the proposed model architecture from the perspective of underlying feature disentanglement [75], [76]. Comprehensively observing the feature distribution of the four major datasets of Vaihingen, Potsdam, LoveDA, and GID (Fig.10), it can be seen that due to the inherent high intra-class variance and cross-scene heterogeneity of remote sensing images, the feature space of the original images presents a highly disordered and interleaved state. When processing these four types of complex scenes, conventional comparison models generally have significant inter-class aliasing phenomena. Especially when facing severe spectral ambiguity and drastic scale changes, the feature extraction mechanism with a single receptive field is difficult to effectively decouple the complex urban scene semantics, resulting in feature clusters of different categories overlapping and having blurred boundaries. As shown in Fig. 11, the feature distribution of DSFANet on the four heterogeneous datasets all shows a significantly improved clustering topology, which not only forms clear decision boundaries, but also highly matches the theoretical expectations of the spatial-frequency dual-domain collaborative architecture proposed in this paper.

E. Accuracy and Efficiency Analysis of Different Models

The comprehensive performance comparison chart shown in Fig. 12 reveals the trade-off relationship between segmentation accuracy (mIoU) and computational overhead (Params/FLOPs) among the models. It can be observed from Fig. 12(a) and 12(b) that while achieving the highest segmentation accuracy, the model parameter count of DSFANet is significantly compressed to the lowest level among all comparison methods, and the computational complexity (FLOPs) is also maintained in a highly competitive lightweight range. This “high accuracy-extremely low overhead” performance confirms the high efficiency of the spatial-frequency dual-domain collaborative architecture in this paper from the dimension of engineering complexity: On the one hand, the spatial LightLKA branch abandons the parameter-expensive conventional convolution. By introducing Dilated Depth-wise Convolution, it successfully constructs a super-large local receptive field for capturing high-frequency geometric edges at a very small parameter cost. On the other hand, the Global Filter module in the frequency domain replaces the compute-intensive dot-product self-attention in traditional Transformers with efficient frequency-domain multiplication operations, achieving the modeling of global long-range topological dependencies with extremely low computational overhead. In addition, combined with the SCAG gating module and the feature fusion mechanism with a strong compression ratio, the network effectively mitigates strips away redundant background noise and channel calculations.

Furthermore, to comprehensively evaluate the model's practical deployment capabilities—particularly when dealing with extremely large remote sensing datasets and real-time processing scenarios—we extended our analysis to empirical hardware efficiency. As illustrated in Fig. 12(c), DSFANet achieves an inference speed of 174.66 FPS. Alongside its rapid runtime, the peak GPU memory usage of DSFANet is constrained to 84.03 MB. This exceptionally low memory footprint is highly advantageous for large-scale earth observation tasks, as it allows for processing significantly larger input tile sizes in a single forward pass, thereby circumventing Out-of-Memory (OOM) errors and minimizing the severe I/O overhead associated with dense image cropping and stitching. In summary, DSFANet does not rely on blindly expanding network depth or channel capacity in exchange for accuracy dividends; instead, it achieves a highly efficient, deployment-friendly remote sensing feature representation through the innovation of underlying dual-domain decoupling and adaptive routing mechanisms.

VI. CONCLUSION

To tackle the formidable challenges of fine-grained segmentation within complex high-resolution remote sensing

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scenarios, this paper proposes a spatial-frequency dual-domain collaborative decoupling network that balances accuracy and efficiency. By innovatively coupling the 2D-FFT based frequency-domain global guidance (with a complexity of $O(Nlog\ N)$) and the spatial-domain lightweight large kernel attention, this model effectively overcomes the inherent defect of deep networks that makes it difficult to simultaneously capture macroscopic long-range topological dependencies and retain microscopic high-frequency geometric boundaries. Furthermore, through the SCAG and linear attention fusion mechanisms, the network achieves adaptive alignment and efficient decoding of features. Experiments on four remote sensing datasets show that compared with the advanced methods investigated, DSFANet achieves the optimal trade-off between accuracy and computational overhead. However, this study still has certain limitations; the adaptive feature extraction ability of the current frequency-domain filtering mechanism for non-stationary extreme geographic textures still has room for optimization, and the model performance still highly relies on high-quality, large-scale, fully supervised annotated data. Future work will be dedicated to introducing the spatial-frequency decoupling theory into semi-supervised/unsupervised domain adaptation frameworks, and expanding it to broader multi-modal Earth observation tasks.

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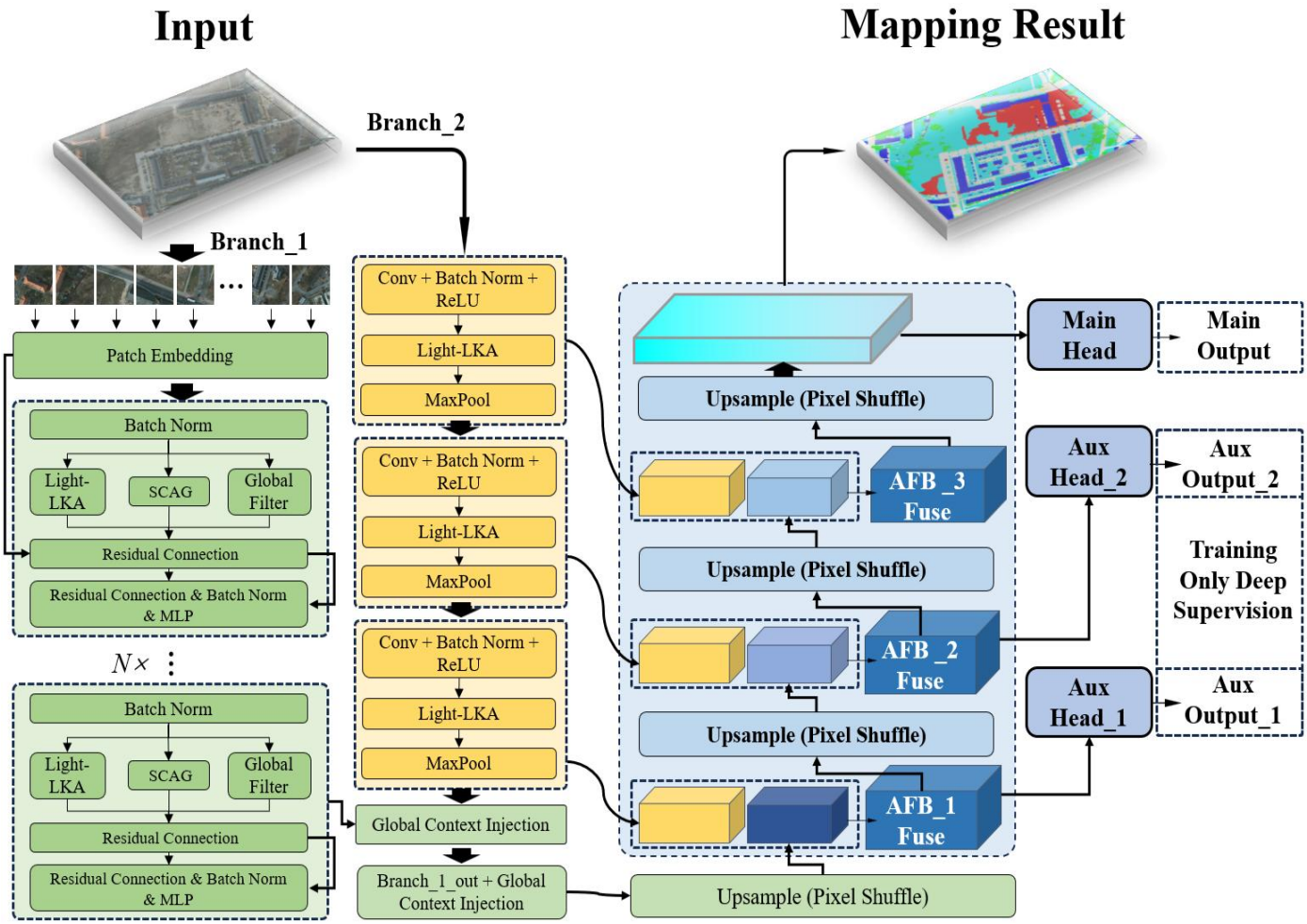


Fig.1. Overview of the proposed mapping framework.

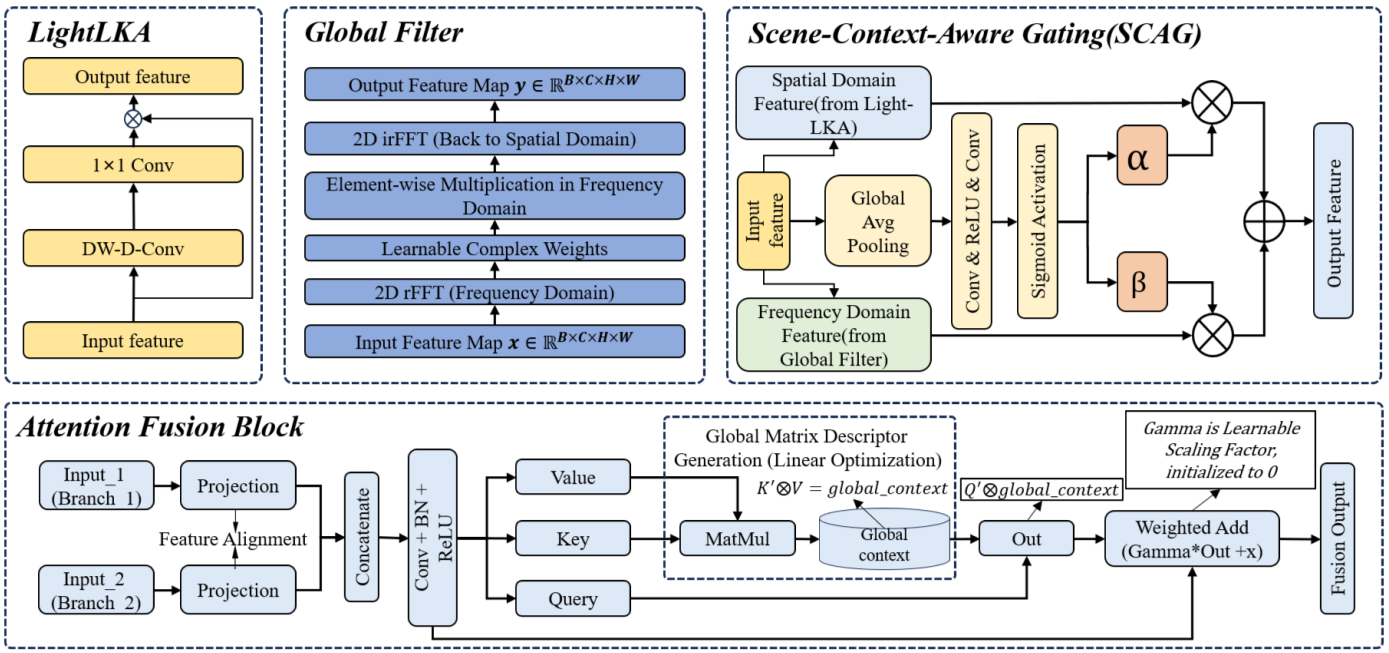


Fig.2. Details of the proposed modules.

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TABLE I
RESULTS OF COMPARATIVE EXPERIMENTS CONDUCTED ON THE VAIHINGEN DATASET

Method	Imp. surf.	Building	Low.veg	Tree	Car	mIoU	mF$_1$	OA	Kappa
ABCNet	85.40(92.13)	85.84(92.38)	74.70(85.52)	81.78(89.98)	71.06(83.08)	79.76%	88.62%	90.40%	0.8718
BANet	78.47(87.94)	78.93(88.23)	63.03(77.33)	74.51(85.40)	52.02(68.44)	69.39%	81.47%	85.24%	0.8023
CMLFormer	80.24(89.04)	81.34(89.71)	63.75(77.86)	75.66(86.15)	60.29(75.23)	72.26%	83.60%	86.48%	0.8191
CMTFNet	85.69(92.80)	86.78(92.35)	72.12(83.87)	81.24(89.65)	**77.09(87.06)**	80.58%	89.15%	90.44%	0.8723
MANet	85.39(92.12)	87.14(93.13)	73.25(84.56)	80.75(89.35)	68.76(81.49)	79.06%	88.13%	90.31%	0.8706
SFFNet	82.10(90.17)	81.27(89.67)	65.36(79.05)	76.82(86.89)	55.15(71.09)	72.14%	83.38%	87.07%	0.8265
AFENet	75.13(85.80)	75.15(85.81)	67.44(80.55)	75.12(85.79)	62.95(77.26)	71.16%	83.04%	84.64%	0.7949
MIFNet	81.85(90.02)	83.79(91.18)	70.08(82.41)	78.05(87.68)	61.63(76.26)	75.08%	85.51%	88.20%	0.8425
SFCVPNet	78.26(87.81)	80.91(89.45)	63.73(77.85)	75.40(85.97)	39.89(57.03)	67.64%	79.62%	85.51%	0.8058
DSFANet	**86.68(92.87)**	**87.23(93.18)**	**76.91(86.95)**	**82.62(90.48)**	74.95(85.68)	**81.68%**	**89.83%**	**91.28%**	**0.8837**

Note: the value outside the parentheses is the per-class IoU (%), and the value inside the parentheses is the per-class F1-score (%). The best results are in bold.

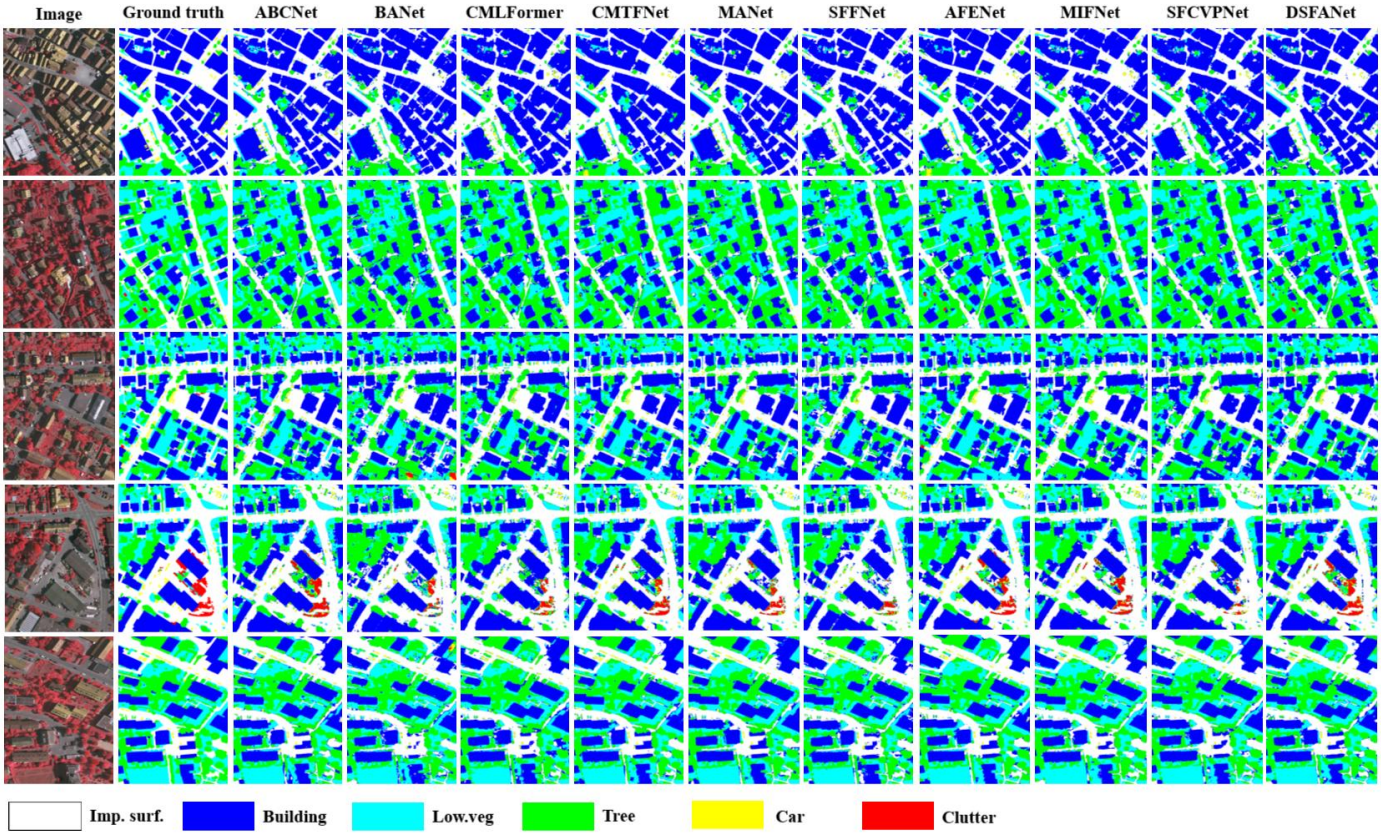


Fig.3. Visualization results of the Vaihingen datasets.

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TABLE II
RESULTS OF COMPARATIVE EXPERIMENTS CONDUCTED ON THE POTSDAM DATASET.

Method	Imp. surf.	Building	Low.veg	Tree	Car	mIoU	mF_1	OA	Kappa
ABCNet	81.27(89.67)	84.47(91.58)	75.56(86.08)	67.17(80.36)	67.30(80.45)	75.15%	85.63%	87.35%	0.8345
BANet	73.50(84.72)	76.60(86.75)	64.67(78.54)	53.19(69.44)	59.72(74.78)	65.53%	78.85%	80.74%	0.7483
CMLFormer	77.43(87.28)	80.14(88.98)	67.48(80.58)	56.81(72.46)	59.42(74.54)	68.26%	80.77%	83.20%	0.7796
CMTFNet	83.05(90.74)	84.82(91.79)	77.36(87.23)	71.07(83.09)	69.71(82.15)	77.20%	87.00%	88.45%	0.8491
MANet	81.90(90.05)	**85.98(92.46)**	75.20(85.84)	69.93(82.31)	68.37(81.22)	76.28%	86.37%	87.93%	0.8424
SFFNet	78.68(88.07)	80.91(89.45)	68.47(81.28)	60.99(75.77)	62.00(76.54)	70.21%	82.22%	84.17%	0.7931
AFENet	82.75(90.56)	85.01(91.90)	76.42(86.63)	68.71(81.45)	70.65(82.80)	76.71%	86.67%	88.04%	0.8441
MIFNet	79.93(88.85)	83.37(90.93)	71.25(83.21)	62.62(77.01)	62.95(77.26)	72.02%	83.45%	85.50%	0.8107
SFCVPNet	62.77(77.13)	66.78(80.08)	53.58(69.77)	41.97(59.12)	37.25(54.28)	52.47%	68.08%	72.05%	0.6340
DSFANet	**85.20(92.01)**	84.35(91.51)	**78.54(87.98)**	**73.98(85.04)**	**72.43(84.01)**	**78.90%**	**88.11%**	**89.31%**	**0.8603**

Note: the value outside the parentheses is the per-class IoU (%), and the value inside the parentheses is the per-class F1-score (%). The best results are in bold.

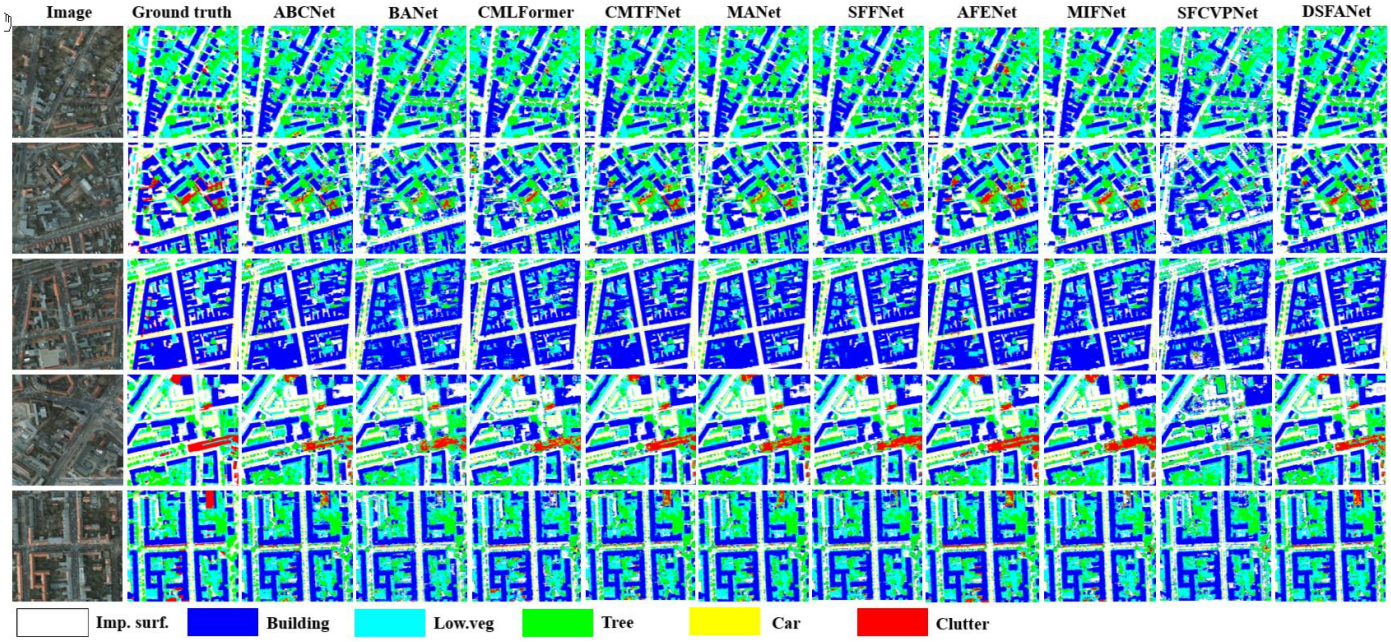


Fig.4. Visualization results of the Potsdam datasets.

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TABLE III
RESULTS OF COMPARATIVE EXPERIMENTS CONDUCTED ON THE LOVEDA DATASET.

Method	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU	mF1	OA	Kappa
ABCNet	56.94 (72.36)	51.81 (68.60)	48.30 (65.05)	**78.59 (88.01)**	51.41 (67.38)	50.86 (65.10)	8.95 (16.13)	49.55%	63.25%	59.18%	0.5095
BANet	54.47 (70.18)	44.57 (62.55)	42.99 (63.01)	75.48 (87.31)	40.15 (51.20)	44.28 (61.21)	9.17 (15.22)	44.44%	58.67%	52.09%	0.4372
CMLFormer	52.94 (70.74)	54.79 (69.62)	48.12 (64.95)	73.34 (82.75)	43.04 (60.24)	44.46 (61.63)	**9.41 (16.34)**	46.59%	60.90%	57.23%	0.4883
CMTFNet	51.81 (67.47)	46.26 (62.19)	42.56 (58.68)	70.40 (82.63)	48.41 (66.12)	48.67 (65.52)	7.67 (13.77)	45.11%	59.48%	54.46%	0.4601
MANet	55.68 (71.06)	46.34 (64.17)	43.82 (60.14)	73.36 (85.59)	**54.89 (69.59)**	48.91 (64.72)	8.70 (14.01)	47.39%	61.33%	55.36%	0.4735
SFFNet	53.69 (69.71)	47.05 (62.41)	46.00 (61.95)	72.47 (84.56)	46.12 (63.16)	**56.56 (72.26)**	8.58 (14.81)	47.21%	61.27%	56.02%	0.4783
AFENet	52.62 (68.65)	54.29 (70.03)	38.15 (55.29)	71.97 (81.38)	44.21 (62.27)	56.49 (72.19)	8.59 (14.82)	46.62%	60.66%	57.81%	0.4921
MIFNet	51.20 (69.12)	48.02 (67.39)	35.07 (54.20)	62.11 (77.02)	31.22 (42.58)	41.98 (55.13)	8.42 (14.54)	39.72%	54.28%	49.21%	0.4024
SFCVPNet	50.51 (67.39)	44.10 (60.20)	46.11 (65.77)	62.64 (75.78)	45.35 (63.15)	41.40 (55.56)	8.00 (13.81)	42.59%	57.38%	52.10%	0.4355
DSFANet	**57.00 (72.61)**	**55.17 (71.94)**	**52.41 (70.03)**	72.04 (81.02)	47.43 (64.05)	54.33 (68.44)	8.88 (16.32)	**49.61%**	**63.49%**	**59.68%**	**0.5161**

Note: the value outside the parentheses is the per-class IoU (%), and the value inside the parentheses is the per-class F1-score (%). The best results are in bold.

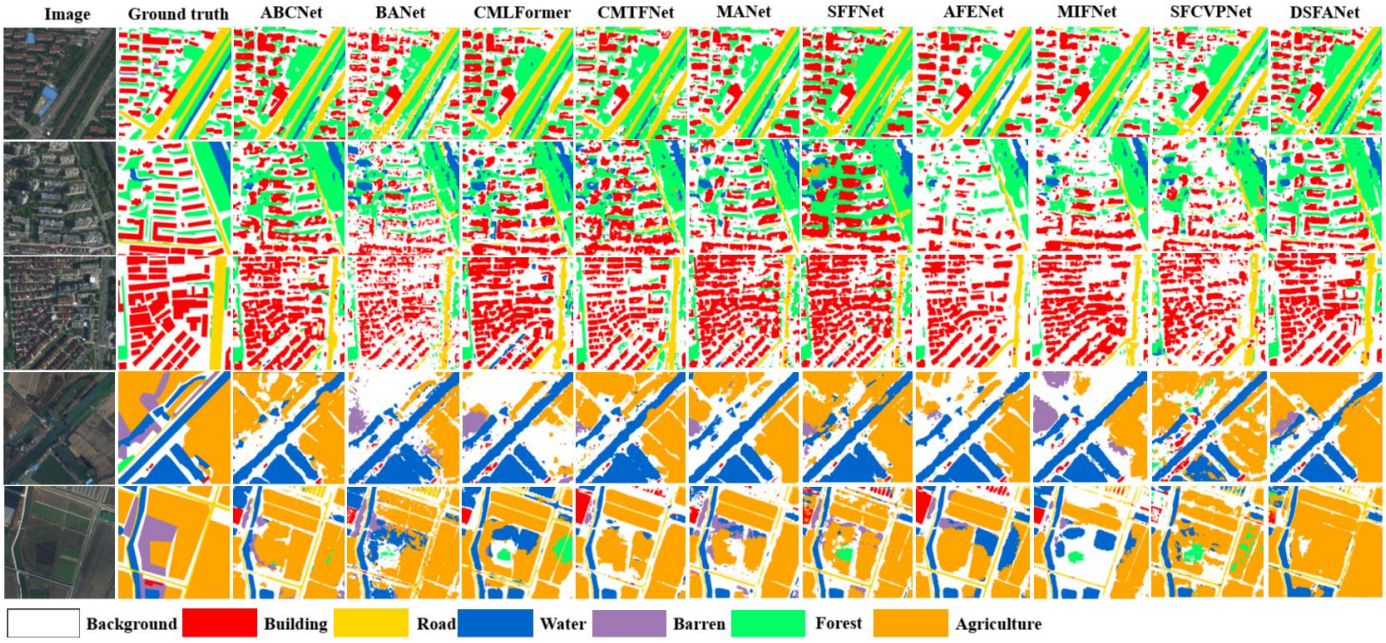


Fig.5. Visualization results of the LoveDA datasets.

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TABLE IV
RESULTS OF COMPARATIVE EXPERIMENTS CONDUCTED ON THE GID DATASET.

Method	Built-up	Farmland	Forest	Meadow	Water	mIoU	mF1	OA	Kappa
ABCNet	68.66(81.42)	69.58(82.06)	59.44(74.56)	20.24(33.67)	89.39(94.40)	61.46%	73.22%	82.98%	0.7327
BANet	64.96(78.76)	66.37(79.79)	62.25(76.74)	19.66(32.86)	88.65(93.98)	60.38%	72.43%	80.88%	0.7005
CMLFormer	67.85(80.85)	68.65(81.41)	60.60(75.47)	4.11(7.90)	88.47(93.88)	57.94%	67.90%	82.25%	0.7209
CMTFNet	73.57(84.77)	75.19(85.84)	63.24(77.48)	15.71(27.20)	92.82(96.28)	64.11%	74.31%	86.28%	0.7843
MANet	**75.45(86.01)**	74.20(84.63)	57.82(75.28)	20.40(34.75)	92.27(95.98)	64.03%	75.33%	85.53%	0.7730
SFFNet	63.86(77.94)	65.26(78.98)	58.29(73.65)	10.86(19.59)	86.73(92.89)	57.00%	68.61%	79.84%	0.6832
AFENet	72.48(82.37)	75.09(85.06)	59.55(75.05)	20.03(33.98)	**93.96(97.35)**	64.22%	74.76%	86.95%	0.7949
MIFNet	71.17(83.16)	72.01(83.73)	**66.36(79.78)**	16.60(28.48)	90.97(95.27)	63.42%	74.08%	84.62%	0.7587
SFCVPNet	48.39(65.22)	45.23(62.28)	23.72(38.34)	0.10(0.20)	67.08(80.30)	36.90%	49.27%	64.29%	0.4387
DSFANet	73.25(84.16)	**76.58(86.74)**	64.10(78.12)	**23.18(36.56)**	91.20(94.76)	**65.66%**	**76.07%**	**87.09%**	**0.7971**

Note: the value outside the parentheses is the per-class IoU (%), and the value inside the parentheses is the per-class F1-score (%). The best results are in bold.

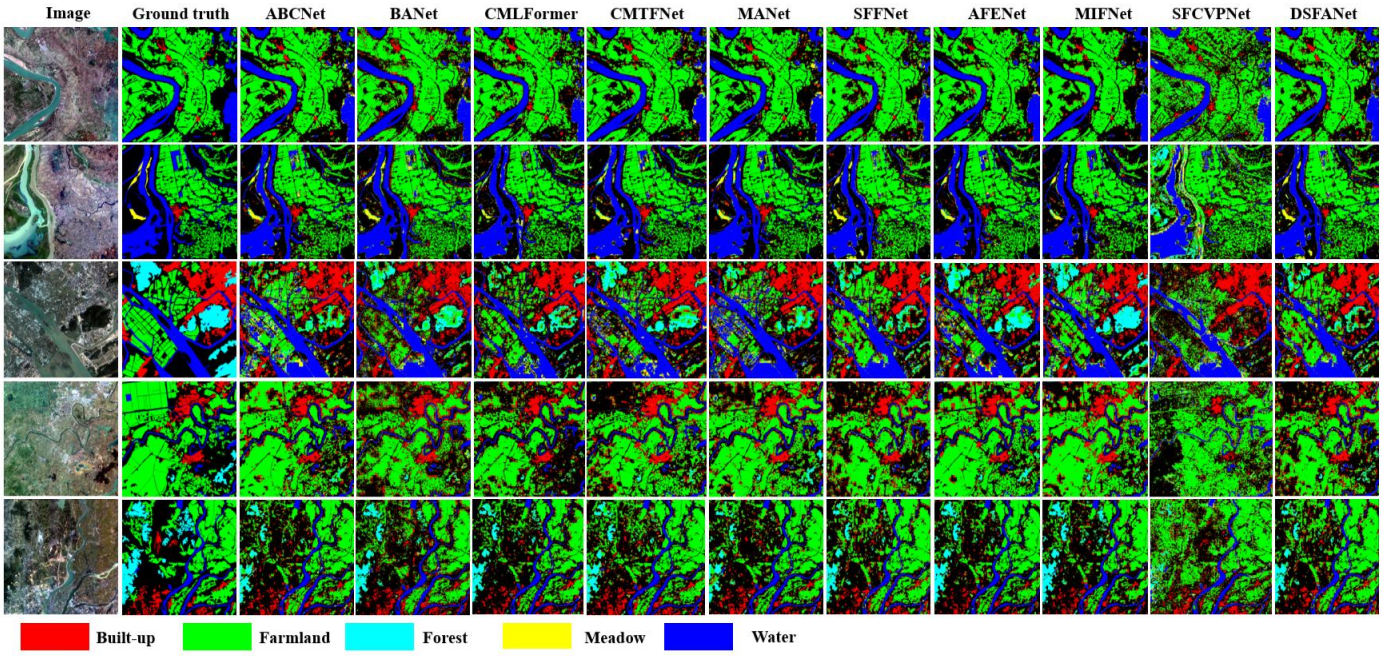


Fig.6. Visualization results of the GID datasets.

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TABLE V
RESULTS OF ABLATION ON THE VAIHINGEN DATASET.

Method	Baseline	LightLKA	Global filter	Element-wise addition	SCAG	AFB	$\mathcal{L}_{ds}$	OA	mF1	mIoU	Kappa
Ablation1	√							77.66%	70.61%	56.37%	0.7000
Ablation2	√	√						82.37%	80.28%	69.25%	0.7634
Ablation3	√		√					85.65%	83.28%	71.83%	0.7926
Ablation4	√	√	√	√				86.42%	84.15%	73.13%	0.8123
Ablation5	√	√	√		√			87.69%	85.54%	75.07%	0.8354
Ablation6	√	√	√		√	√		90.41%	88.47%	79.55%	0.8718
Ablation7	√	√	√		√	√	√	**91.28%**	**89.83%**	**81.68%**	**0.8837**

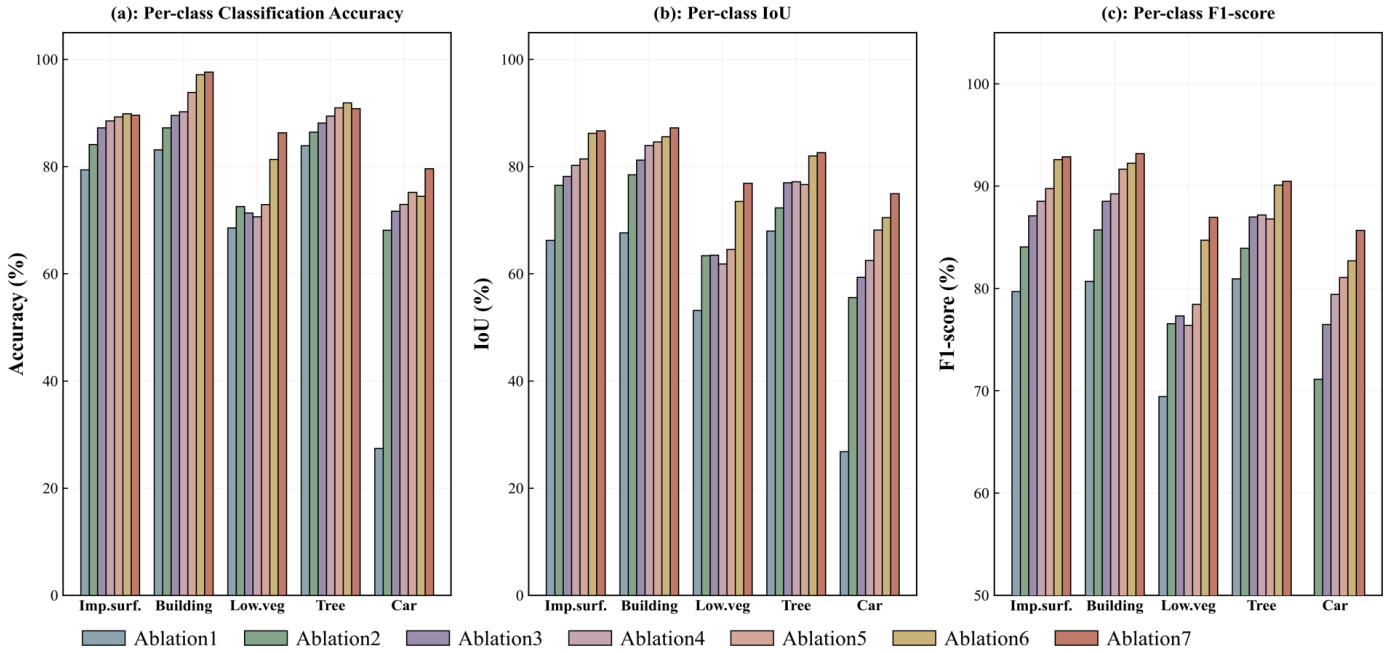


Fig.7. Ablation study results: (a): Per-class Classification Accuracy, (b): Per-class IoU, (c): Per-class F1-score.

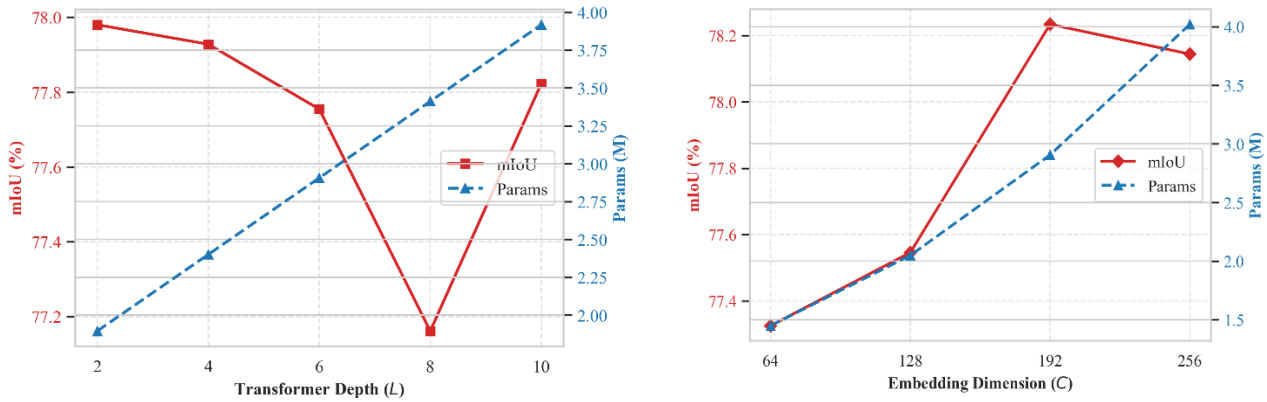


Fig.8. Parameter sensitivity analysis of transformer depth (L) and embedding dimension (C).

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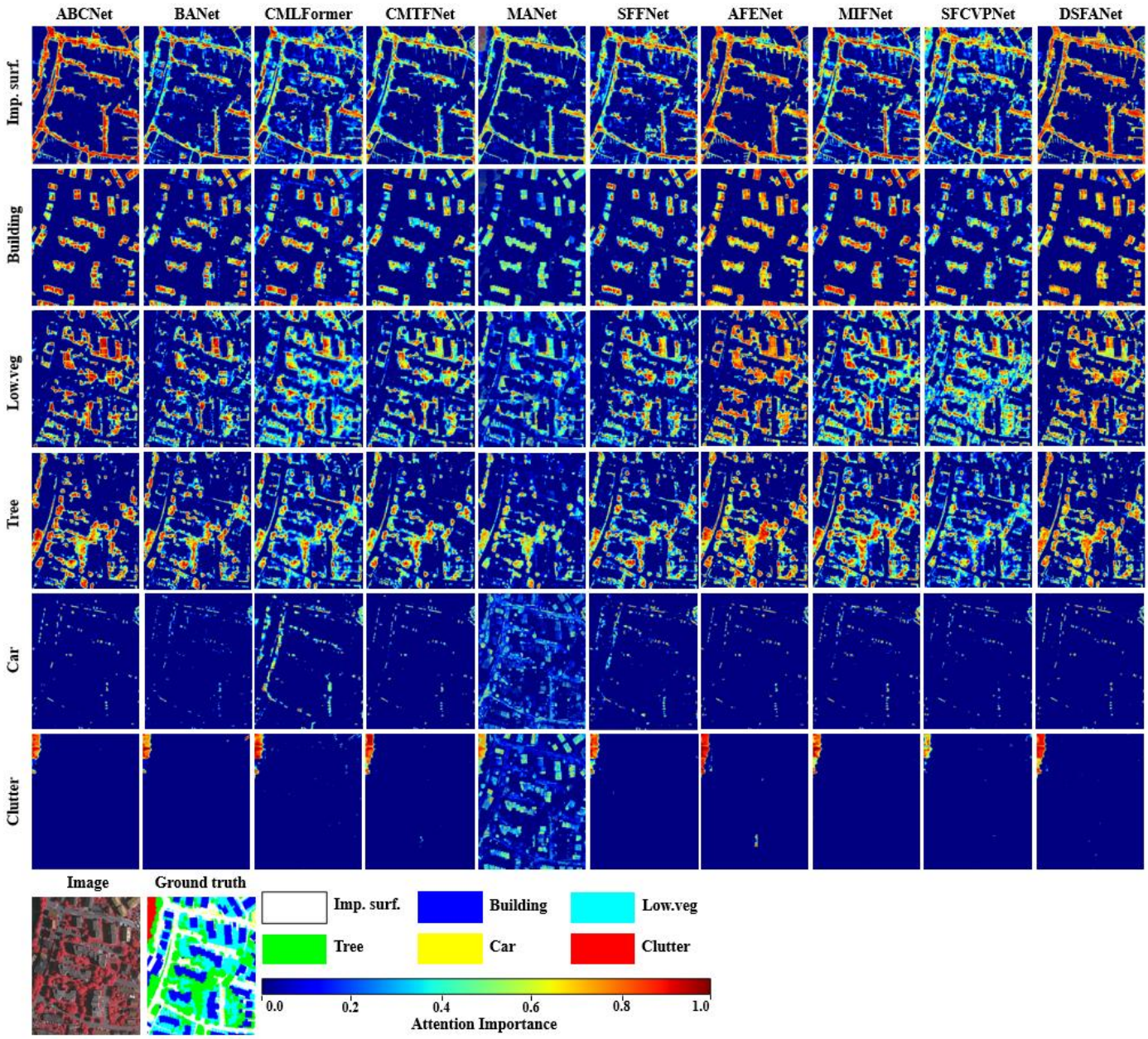


Fig.9. Grad-CAM visualization results.

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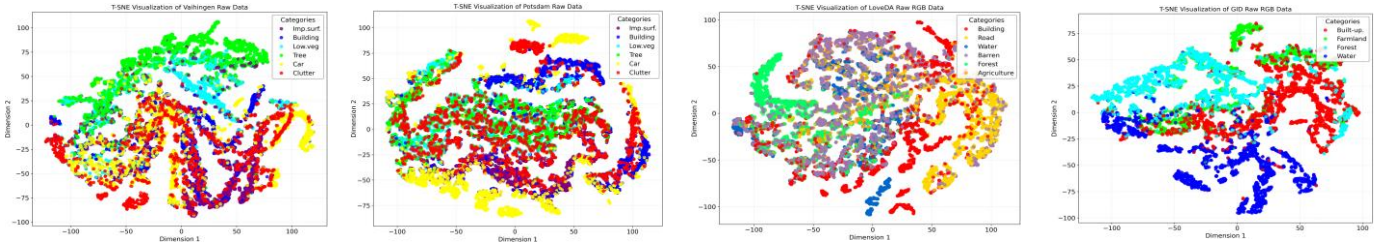
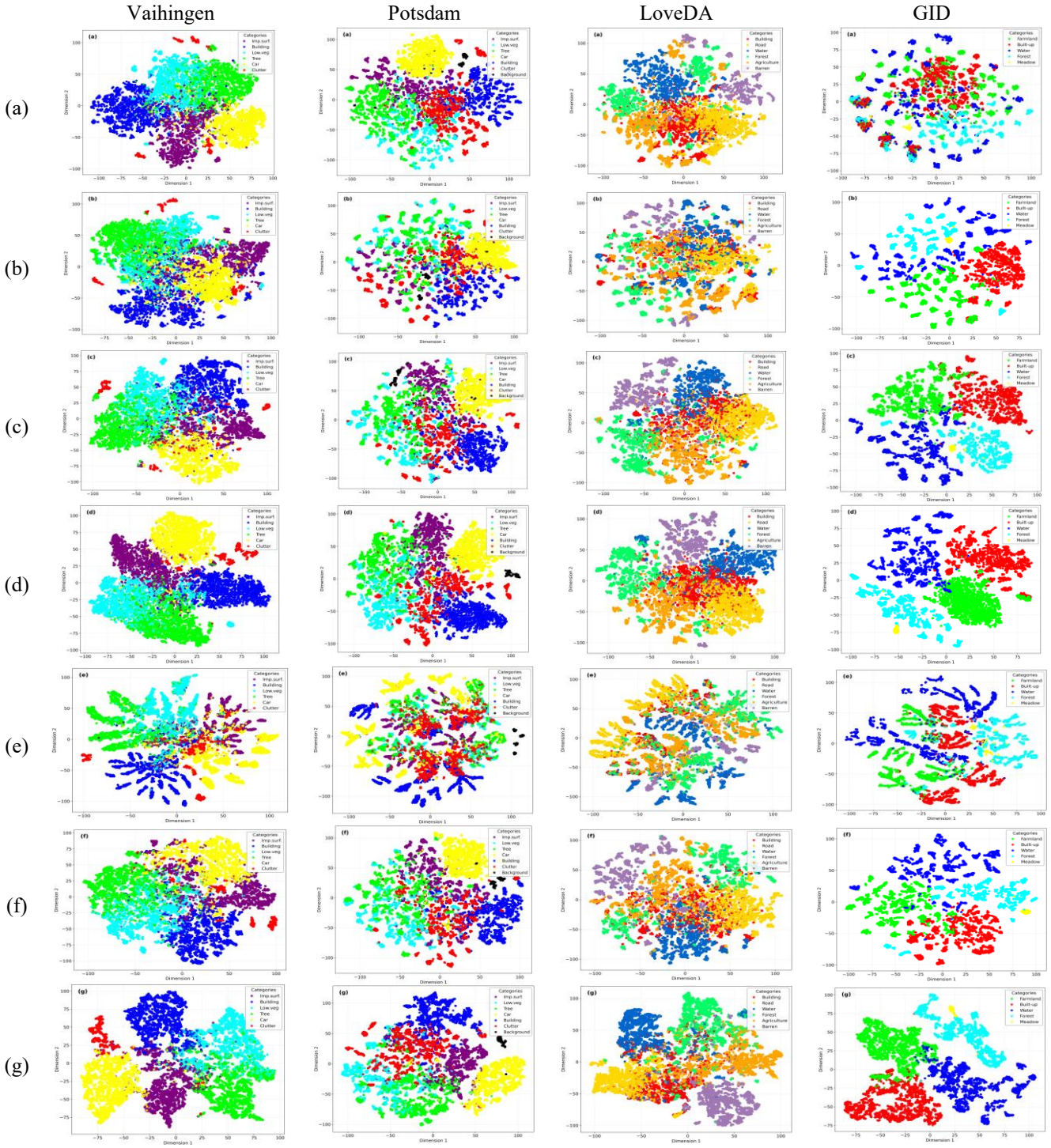


Fig.10. t-SNE visualizations of Vaihingen, Potsdam, LoveDA, GID raw data



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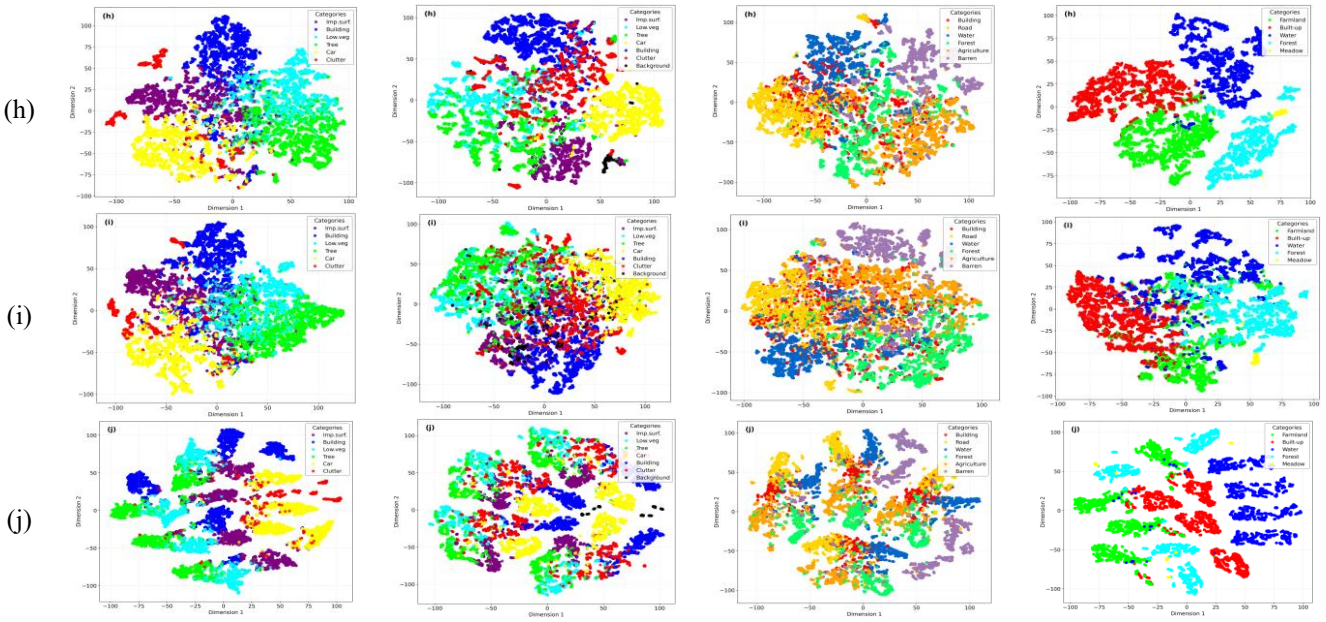


Fig.11. Comparative t-SNE visualizations of deep feature spaces across different models, (a) ABCNet, (b) BANet, (c) CMLFormer, (d) CMTFNet, (e) MANet, (f) SFFNet, (g) AFENet, (h) MIFNet, (i) SFCVPNet, (j) DSFANet.

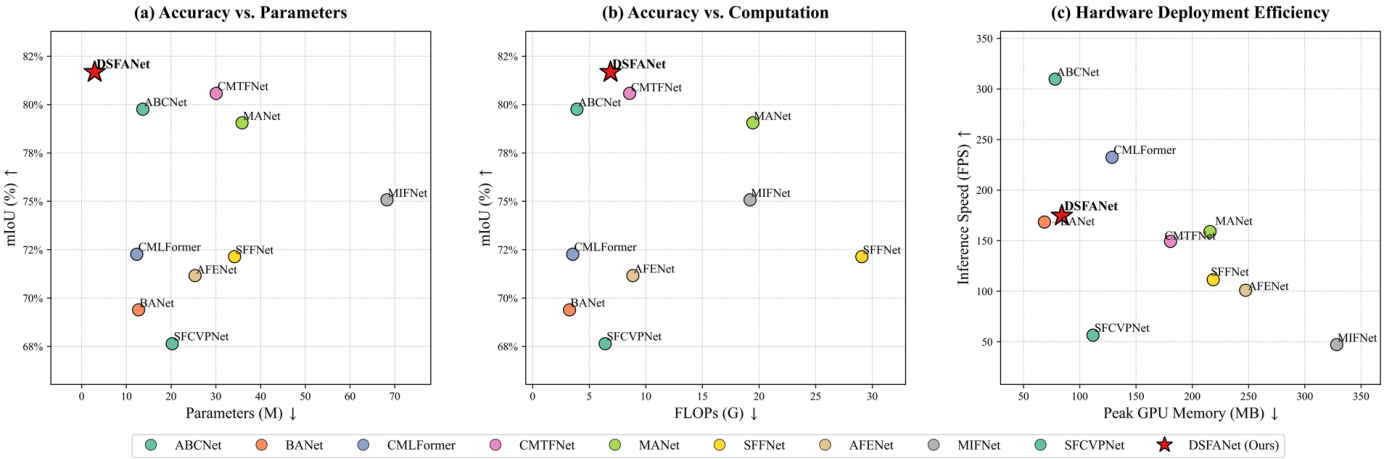


Fig.12. Accuracy and efficiency analysis of different models.

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